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Separation of Preadolescents from Infant Rhesus Monkeys*

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ABSTRACT. Each of eight infant rhesus monkeys was paired with a preadolescent conspecific for two months and then separated. Four of the infants were mother-reared and four were isolate-reared. Separation responses were compared with data from preseparation and reunion phases of the study for all pairs. The results indicate that (1) although males interact with infants in a parental fashion, preadolescent females show a greater capacity for parental behavior, (2) both preadolescents and normal (mother-reared) infants contribute to the development of a social bond, (3) isolate infants contributed little to the development of a social bond and were relatively less-valued as social partners by their preadolescent cagemates.

INTRODUCTION

Many studies of nonhuman primates and other mammals have shown that behaviors important to normal adult social and parental functions are dependent upon prior social experience. This experience must meet certain minimal requirements in order to prepare the animal for a successful adult life. The early work of HARLOW and his colleagues, for example, revealed that social deprivation of an extreme sort (social isolation) creates animals which are unable to copulate efficiently (cf. HARLOW & HARLOW, 1962). Other behavioral deficiencies in these animals are apparent when social contact with a normal animal is attempted. The social isolate has become a working model for researchers interested in behavioral pathology.

In our research at the California Primate Research Center we have been concerned with the development of attachment in various age and sex pairs of rhesus monkeys (*Macaca mulatta*). The strategy utilized in our research generally calls for the physical separation of the pair after the animals have lived together for various specified periods of time (see, for example, ERWIN, MOBALDI, & MICHELL, 1971; ERWIN, BRANDT, & MITCHELL, 1973). Separation of monkeys and other mammals has been investigated intensively in several different laboratories (cf. SCOTT & BRONSON, 1964; ROSENBLUM & KAUFMAN, 1968; HINDE & SPENCER-BOOTH, 1971; SCHLOTTMANN & SEAY, 1972; MCKINNEY, SUOMI, & HARLOW, 1972). The present research report describes behavior during and fo lowing the separation of infants from preadolescents after the two age groups had lived in heterogeneous dyads for a 2-month period. This study is the second in a general investigation of the nature of attachment between infants and preadolescents (see BRANDT & MITCHELL, 1973). Further aims of this

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research have been to gather data germane to the following topics: (a) the development of maternal and paternal behavior before puberty in preadolescents, (b) general sex differences in infants and preadolescents, (c) general age differences between infants and preadolescents, and (d) the effects of early social isolation on infant-preadolescent interactions.

METHOD

SUBJECTS

Each of eight infant rhesus monkeys was paired with a preadolescent (30 months old) conspecific. Four of the infants, two males and two females, had been raised alone with their mothers and weaned at seven months of age. These four infants were matched for age and sex with other infants who had been raised in social isolation for the first seven months of life (see BAYSINGER, BRANDT, & MITCHELL, 1972). All of the preadolescents were reared with mothers from birth and had interacted with peers from the seventh month of life. This peer experience included one prior separation from an age mate (ERWIN, MOBALDI, & MITCHELL, 1971). In the present study each preadolescent had lived with his or her infant for two months prior to the separation (BRANDT & MITCHELL, 1973). The preadolescent/infant pairings were made so that both rearing experience and sex of infant were balanced across sex of preadolescent. All possible sex combinations were formed: male preadolescent with female infant, female preadolescent with male infant, male preadolescent with male infant, and female preadolescent with female infant. Table 1 gives more detailed information concerning subjects.

APPARATUS

The apparatus is described fully in ERWIN, MOBALDI, and MITCHELL (1971). Briefly each infant-preadolescent pair was housed in a $36 \times 36 \times 36$ inch home cage with auditory and visual contact with the other pairs. Separated subjects were housed in

Table 1. Detailed information concerning subjects.

Pair	No.	Subject	Sex	Birthdate	Experience with mother	Age upon pairing
1	5875	juvenile	M	6-10-68		30 months
	6752	inf. control	M	3-25-70	7 months	8 months
2	5870	juvenile	M	6-4-68		30 months
	6755	inf. isolate	M	3-27-70	0 month	8 months
3	6001	juvenile	M	7-23-68		31 months
	6809	inf. control	F	6-22-70	7 months	8 months
4	5995	juvenile	M	7-15-68		31 months
	6808	inf. isolate	M	6-22-70	0 month	8 months
5	6347	juvenile	F	9-20-68		30 months
	6807	inf. control	M	6-22-70	7 months	8 months
6	6243	juvenile	F	9-14-68		30 months
	6811	inf. isolate	M	6-24-70	0 month	8 months
7	6134	juvenile	F	8-28-68		31 months
	6829	inf. control	F	7-19-70	7 months	8 months
8	6119	juvenile	F	8-21-68		31 months
	6833	inf. isolate	F	7-20-70	0 month	8 months

experimental rooms containing a $36 \times 36 \times 72$ inch test cage and a clock counter (ODOM, MITCHELL, & LINDBURG, 1970). All observations were taken in the test cage by observers who were in full view of the monkeys.

PROCEDURE

Prior to data collection, each infant-preadolescent pair was put into the test cage for a one-day adaptation period. A six-day study followed: two days of preseparation, two days of separation, and a final two days of postseparation (reunion) (see ERWIN, MOBALDI, & MITCHELL, 1971; ERWIN, BRANDT, & MITCHELL, 1963 for the rationale behind the two-day separation). Observations were made simultaneously for each matched set of isolate and control infants and their respective cage-mates (preadolescents).

During the study period, 50 behaviors were scored. For convenience in scoring, these behaviors were categorized as follows: A; vocalization and aggression, B; other-animal directed behavior, C; exploration, D; visual orientation/other and, E; isolate behavior. Table 2 lists these behaviors together with brief definitions. Observers recorded frequencies and durations of behaviors using a clock-counter (ODOM, MITCHELL, & LINDBURG, 1970). A 2.5 minute time sample was obtained for each behavior category per subject each day. By this procedure, the individual behaviors in each category (e.g. vocalizations and aggression) are represented by a 2.5 minute time sample. Testing took place in the afternoon. The order of testing for the behavior categories remained the same each day.

Two days of scoring baseline behavior (preseparation) was followed by initiation of the separation procedure. This was accomplished by sliding an opaque panel between the two animals. Scoring for each subject began at the moment the partition was intact. The partition remained in place for two days and the behaviors were scored once on day three and day four. The partition was removed on the fifth experimental day. The fifth and sixth days made up the reunion phase of the study. The fifth day of scoring began at the moment the partition separating the animals was removed.

STATISTICS

In the initial stages of statistical inquiry, repeated-measures analyses of variance were utilized with the factors being as follows: infant rearing (isolate versus control), sex of infant, sex of preadolescent, study phase (presepartion, separation or postseparation), and sessions (day 1 or day 2 within phase).

RESULTS

CONTROL INFANTS AND PREADOLESCENTS

Vocalizations

Normal infants vocalized much more often than did the preadolescents from whom they were separated (see Table 3). While preadolescents were never heard to screech, girn, or gecker, two of the four control infants emitted all of these vocalizations and the other two controls exhibited two of the three types of vocalizations. Both infants

Table 2. Brief definitions of behaviors scored.

A. Vocalization/aggression
1. *Coo*: soft "ooo" vocalization.
2. *Screech*: high pitched scream.
3. *Girn*: nasal grunting vocalization.
4. *Gecker*: high pitched broken staccatto sound.
5. *Bark*: low pitched explosive sound.
6. *Threaten juvenile/infant*: an open-mouthed facial expression.
7. *Clasp-pull*: grasping and pulling other's fur.
8. *Bite*: common usage.
9. *Cage shake*: shaking the cage.
10. *Yawn*: common usage.

B. *Other-animal directed behavior*
11. *Present*: orienting hindquarters toward other animal.
12. *Mount*: climbing upon other animal, feet clasping calves and hands grasping loins.
13. *Thrust*: pelvic thrusting.
14. *Make contact*: establish any body contact with other animal.
15. *Break contact*: breaking body contact with other animal.
16. *Manual explore*: manipulating other: excludes grooming (see No. 20).
17. *Oral-olfactory explore*: oral contact with or smelling of other; excludes oral aggression (biting).
18. *Lipsmack*: friendly facial expression.
19. *Fear grimace*: fearful facial expression.
20. *Groom*: picking through fur of other.

C. *Exploration*
21. *Locomotion on floor*: walking or running on cage floor.
22. *Locomotion on wall*: climbing on walls or ceiling.
23. *Jump*: common usage.
24. *Stereotyped movement*: repetitive locomotion (e.g., pacing).
25. *Manual explore environment*: common usage.
26. *Oral-olfactory explore environment*: common usage.
27. *Manual explore self*: common usage.
28. *Oral olfactory explore self*: common usage.
29. *Self-groom*: see No. 20.
30. *Play*: highly active jumping and/or running with orientation to and manipulation of the other animal.

D. *Visual orientation/other*
31. *Self*: eyes oriented toward own body.
32. *Infant/juvenile*: eyes oriented toward other animal.
33. *Observer*: eyes oriented toward observer.
34. *Environment*: eyes oriented toward environment, excluding self, other, observer.
35. *Threaten observer*: see No. 6.
36. *Lipsmack to observer*: see No. 8.
37. *Fear grimace to observer*: see No. 19.
38. *Convulsive jerk*: infant stiffens and jerks body.
39. *Open-mouth smacking*: slow smacking of lips.
40. *Ventral contact*: ventral sides of the two animals in cntact.
41. *Nonventral contact*: any body contact excluding No. 40.
42. *Proximity*: animals in same eighth of cage but not in body contact.

E. *Isolate behaviors*
43. *Self-threat*: see No. 6; directed toward self.
44. *Head slap*: hitting own head.
45. *Bizarre movement*: unusual movement.
46. *Abnormal posture*: unusual posture.
47. *Finger suck*: common usage.
48. *Toe suck*: common usage.
49. *Self-clasp*: holding on to part of own body.
50. *Self-bite*: common usage.
51. *Autoerotic*: masturbation.

Table 3. Infants compared with preadolescents: Mean frequencies and/or mean durations in 1/100 minutes.*

Behavior	Measure	Infants	Preadolescents	*P*
Coo	freq.	15.46	3.50	.05
Screech	freq.	3.71	0.00	.10
Girn	freq.	11.54	0.00	.10
Gecker	freq.	1.46	0.00	.10
Locomotion on the wall	dur.	19.13	5.88	.025
Stereotypy	dur.	38.04	84.21	.10
Visual: other animal	freq.	11.04	6.04	.10
Visual: environment	freq.	19.13	11.21	.10
Oral explore self	dur.	2.88	0.29	.10
Break contact	freq.	0.94	2.81	.10

*The number in Tables 3–7 represent transformations of the data from seconds to hundredths of one minute. Five seconds would therefore equal 8.33 units (or approximately .08).

and preadolescents cooed, but infants cooed significantly more often than preadolescents. Screeches and barks were few in number during preseparation and reunion but increased during separation (see Table 4). The duration of coo decreased over the two sessions of each separation phase for all infants, but for control infants the cumulative duration of coo increased over the 2 days of separation (although coos were not significantly more frequent) and remained high during the first session of reunion (see Table 4). The infants as a group cooed longest (cumulative duration of coo) during the second day of the separation phase and on the first day of reunion.

Activity

Infants climbed on the walls of the test cage more than did preadolescents (see Table 3). A longer total duration of stereotyped movement was exhibited by preadolescents as compared to infants (see Table 3). Preseparation and reunion levels did not differ significantly, but stereotypy increased during separation (see Table 4). This increase was primarily in the infants (see Table 4). Oral exploration of the

Table 4. Separation and reunion effects for preadolescents and control infants combined: Mean frequencies and/or mean durations in 1/100 minutes.

Behavior	Measure		Presep.		Sep.		Reunion		*P.*
Screech	freq.		0.81		4.25		0.50		.001
Bark	freq.		0.38		11.38		0.63		.05
Stereotypy	freq.		2.63		7.13		2.31		.01
Oral explore: environment	freq.		2.38		0.94		4.94		.005
Visual: observer	freq.		8.00		10.06		4.88		.05
Visual: environment	freq.		19.31		10.05		15.69		.025
Stereotypy[1]	freq.	control infants	0.13		8.13		1.75		.10
		preadolescents	5.13		6.13		2.88		
			day 1	day 2	day 1	day 2	day 1	day 2	
Coo[2]	dur.	control infants	16.50	4.50	18.50	38.50	44.50	10.50	.10
		preadolescents	10.50	1.00	13.00	7.25	3.50	2.75	

1) The data presented here are broken down for infants and preadolescents in order to indicate the variability. The significant effect represents an age by stage interaction. 2) These data also represent some interesting differences between infants and preadolescents. The striking increase in infant coos at reunion is no doubt related to the relative indifference of the predolescent partner.

environment decreased during separation while levels during reunion exceeded those attained during preseparation. Both of these differences were significant (see Table 4).

Visual Orientation

Infants tended to look more frequently at the environment and at their cage-mates than did the preadolescents, although the respective durations of looking were not significantly different (see Table 3). Frequency of looking at the observer decreased during reunion, whereas frequency of looking at the environment decreased during separation but returned to near preseparation levels during reunion (see Table 4).

Self-directed Behavior

Infants showed a higher cumulative duration of oral exploration of the self than did preadolescents (Table 3).

Other Animal Directed Behaviors

Only the preseparation and reunion levels were used to calculate the figure for interactive behaviors since interactive behaviors could not occur during separation. The preadolescents broke contact with their infants more frequently than the infant broke contact with them (see Table 3).

CONTROL INFANTS AND ISOLATE INFANTS

Vocalizations

Control infants tended to coo more often than isolates (Table 6). Cumulative screech duration was by far the longest in control infants paired with males. Control female infants girned much longer than did the other infants.

Activity

Control infants paired with males exhibited the lowest frequencies of movement on the cage floor and male infants moved on the cage floor twice as long as did female infants. Control infants climbed on the cage wall more often than did isolate infants (see Table 6). Isolate females and control males displayed more jumping than the rest of the infants. As one might expect, stereotyped movement was most frequent during separation.

Control males manually explored the environment most often and control females exhibited the lowest levels of this behavior. Oral exploration also occurred most

Table 5. Visual orientation data summarized by age: Mean frequencies and/or mean durations in 1/100 minutes.

Behavior	Measure	Infant	Preadolescent	*P*
Visual orient:	freq.	0.71	0.42	
to self	dur.	2.00	1.96	
Visual orient:	freq.	11.04	6.04	.10
to other animal	dur.	31.42	14.71	
Visual orient:	freq.	9.58	5.71	
to observer	dur.	17.08	11.38	
Visual orient:	freq.	19.13	11.21	
to environment	dur.	200.25	218.00	
Visual orient:	freq.	40.46	23.38	
total	dur.	250.75	246.05	

Table 6. Behavior of infants: Mean frequencies and/or mean durations in 1/100 minutes.

Behavior	Measure	Isolate	Control	*P*
Coo	freq.	3.13	15.46	.10
Locomotion on the wall	freq.	2.54	7.21	.025
Look at self	freq.	1.83	0.71	.10
Abnormal posture	freq.	2.54	0.00	.05
Bizarre movements	dur.	8.85	0.00	.10
Toe sucking	freq.	0.33	0.00	.05
Self bite	freq.	0 21	0.00	.01
Masturbation	dur.	0.84	0 00	.01

often in control males and was generally depressed during separation as compared to the other two phases.

Visual Orientation

Overall frequency of looking at the self was low, but isolates looked at themselves more frequently than did controls (Table 6), especially during separation. Frequency of looking at the preadolescent cage-mate during reunion was almost identical to the preseparation levels, but duration of looking was slightly higher in reunion than in preseparation. Looking at the environment occurred half as frequently in separation as during preseparation or reunion.

Self-directed and Abnormal Behaviors

Female isolates touched their own bodies more frequently than did the other infants. Only the isolates exhibited abnormal postures (Table 6) and these were longest during separation. Only the isolates (and especially the male isolates) exhibited bizarre movements, toe sucking, self-biting, and masturbation (Table 6).

Preadolescent-directed Behaviors

The frequencies of infants making and breaking contact with their preadolescents were low overall. Male infants tended to break contact more often than females. None of the infants threatened, clasp-pulled, bit, mounted, thrusted, or orally explored the preadolescents. Behaviors rarely seen included manual exploration, grooming, play, and present.

Behavior of Preadolescents

Vocalizations

The preadolescents as a group cooed most often during separation and least often during reunion. However, it was the preadolescents paired with infants of the opposite sex who showed the largest increase in coo-vocalizing during separation (see Table 7, $p<.10$). Preadolescents paired with infants of the same sex cooed approximately the same amount of time during separation as during preseparation. Preadolescents with female infants showed an increase in barking during separation. Other vocalizations were emitted too infrequently by the preadolescents to warrant statistical treatment.

Activity

Preadolescents paired with infants of the opposite sex showed a decrease in frequency of normal locomotion on the floor of the cage during separation. Recovery to

Table 7. Behaviors of preadolescents: Mean frequencies and mean durations in 1/100 minutes.

A.	Coo vocalizations (Duration)			
	Dyad	Preseparation	Separation	Reunion
	Male with male infant	3.25	4.00	3.00
	Male with female infant	1.50	11.25	1.50
	Female with male infant	0.75	10.75	2 00
	Female with female infant	11 50	14 00	5 50
B.	Locomotion on floor (Frequency)			
	Dyad	Preseparation	Separation	Reunion
	Male with male infant	3.75	9.25	6.00
	Male with female infant	8.25	2.00	8.00
	Female with male infant	19.50	4.00	16.00
	Female with female infant	3.75	7.25	9.25
C.	Stereotyped movements (Duration)			
	Dyad	Preseparation	Separation	Reunion
	Male with male infant	70.25	103.25	106.50
	Male with female infant	37.75	172.25	56.50
	Female with male infant	8.50	176.00	34.50
	Female with female infant	146.00	64.25	39.75
D.	Manual exploration of environment (Duration)			
	Dyad	Preseparation	Separation	Reunion
	Male with male infant	0.75	2 25	3.75
	Male with female infant	18.25	3.25	26.75
	Female with male infant	20.00	2.75	9.50
	Female with female infant	5.25	7.25	3.25
E.	Visual orientation (Duration)			
	Dyad	Preseparation	Separation	Reunion
	Male with male infant	6 25	not scored	11.00
	Male with female infant	17.00	not scored	22.25
	Female with male infant	47.00	not scored	21.00
	Female with female infant	12 00	not scored	10.50
F.	Make and Break Contact (Frequency)			
	Dyad	Preseparation	Separation	Reunion
	Preadolescents with isolate infants	2.38	—	2.38
	Preadolescents with control infants	7.13	—	2.25

preseparation levels occurred during reunion. Preadolescents with infants of the same sex, on the other hand, showed an increase in locomotion during separation (see Table 7, $p<.025$) which remained at a high level during reunion. The duration data were similar to the frequency data. Preadolescents with infants of the opposite sex showed the lowest frequencies of stereotyped movements, but they also displayed the most dramatic increases in duration of stereotypies during the separation phase (Table 7, $p.<.01$). Stereotypy decreased during reunion but still exceeded preseparation levels in most cases.

Overall, male preadolescents with control infants and female preadolescents with isolate infants showed less cumulative duration of stereotypy than did male preadolescents with isolates and females with controls. Preadolescents, both male and female, with control infants exhibited longer stereotyped movement during separation than during preseparation, and showed a shorter duration of such movements during reunion than during preseparation. Female preadolescents with isolate infants showed

a rather constant duration of stereotypy throughout the study, with a slight decrease at reunion. Male preadolescents with isolates exhibited considerable variability.

Preadolescents paired with infants of the opposite sex manually explored the environment longer than pairs of the same sex. Preadolescents in heterosexual pairs decreased in both the frequency and duration of this behavior at separation but increased again at reunion. In homosexual pairs, the preadolescents actually increased in manual exploration from preseparation to separation (see Table 7, $p<.005$).

Visual Orientation

Preadolescents with male infants looked at themselves least during separation while those with females looked most during separation. Preadolescents looked at heterosexual cagemates significantly more often and for a longer total period of time than they looked at infants of the same sex. Looking at the infants could not occur during separation and there were no consistent differences between reunion and preseparation. Male preadolescents, however, looked slightly more often (and for longer periods of time) during reunion, expecially at female infants (see Table 7, $p.<.05$).

Frequency and cumulative durations of looking at the observer remained constant over preseparation and separation, but decreased during reunion. Overall visual orientation to the environment decreased in frequency from preseparation to separation, but increased slightly at reunion. Females paired with male infants exhibited a very high level of looking at the environment at preseparation with a sharp decrease at separation. The reunion level exceeded that of the separation level but rose to only half of the initial figure for preseparation.

Infant-directed Behaviors

Threats by preadolescents toward infants were not frequent. Mounts and thrusts were also rare, and were attempted only by male preadolescents. Thrusts occurred both during preseparation and reunion for male-female pairs, but only during reunion for male-male pairs.

Preadolescents with isolates made and broke contact equally as often during preseparation and reunion, but preadolescents with controls made and broke contact significantly more often during preseparation than during reunion (Table 7, $p<.025$). Total contact time did *not* increase at reunion as compared to preseparation but the frequency of leaving and returning diminished at reunion if the infant was a control.

Summary of Results

Both the preadolescents and the infants were disturbed at separation. While the preadolescents displayed more stereotyped movements than did the infants, the infant controls showed a greater increase in stereotypy during separation. The control infants vocalized more, climbed on the walls of the test cage more, and displayed more social and environmental looking, and orally explored themselves more than did the preadolescents.

Control infants, in comparison with isolate-reared infants, vocalized more, screeched more in the presence of larger males, girned more (if they were females), and were

more active (if they were males) in the presence of larger females. The isolates could be best characterized as being more self-directed than the controls, and their self-directed behaviors increased during separation.

Perhaps most interesting of all was the finding that the preadolescents had apparently built up a different quality and intensity of attachments for infants of the opposite sex than for infants of the same sex. Their protest as indicated by distress calls (cooing) and stereotyped movements, increased markedly when separated from infants of unlike sex, and this was especially so for pairings with control infants.[3)]

DISCUSSION

PRE-PARENTAL BEHAVIOR

Field researchers (cf. LANCASTER, 1971) have observed play and, occasionally, agonistic interactions between infants and preadolescent conspecifics. Siblings are particularly likely to spend time interacting together, and these interactions may contribute to the development of appropriate social behaviors. Much of the behavior which is directed by an older animal to a younger one may be categorized as pre-parental behavior, although the essential caretaker-role is rarely assumed prior to puberty. In the present experiment, females were more parental than males. Interestingly enough, males seem to have the potential for parental behavior even though (at least for the rhesus monkey) an adult male's parental behavior is infrequent under natural free-ranging conditions. Still, preadolescent females usually show a greater capacity for parental activity than do males, and this sex difference appears under laboratory conditions (CHAMOVE, HARLOW, & MITCHELL, 1967). It is important to note here that the form that this behavior can assume in the field and under laboratory conditions is extremely variable. Males and females, however, are very different in the way they behave toward an infant conspecific, whatever the environment.

Other members of our research team are presently engaged in work which examines the potential for adult paternal behavior. Preliminary results have been described elsewhere (REDICAN & MITCHELL, 1973). Under the proper circumstances, it now seems clear that a strong bond between an infant and an adult male is possible in the rhesus, with the adult male exhibiting a considerable amount of parental behavior.

Our separation paradigm provides us with some information about the relative strength of attachments according to the constituency of a given dyad. However, the development of parental behavior in preadolescent males and females is more clearly accessible in the data of the long-term attachment process. For detailed information on the attachments of the present dyads prior to the start of this experiment the reader is referred to our previously published report (BRANDT & MITCHELL, 1973). These laboratory studies have allowed us to observe pre-parental behavior development and, further, to specify some of the variables which affect it.

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3) Additional descriptive data on each dyad and additional statistical results may be obtained from the authors upon request.

Infants Compared with Preadolescents

The results clearly indicate an age difference with respect to severity of disturbance at separation with the younger animals displaying a much higher level of disturbance (as indicated by a number of criteria) than their preadolescent counterparts. From our data it appears that the younger animals oriented toward the older animals to a greater extent than *vice versa*. The infant is of course expected to show high levels of orientation to an older animal since it would normally spend most of its active time with its mother. Our preadolescents, on the other hand, were relatively inexperienced in child-rearing behaviors and did not behave toward the infants exactly as a mother would. Data obtained from the attachment phase, however (see BRANDT & MITCHELL, 1973), demonstrate that the preadolescents, as well as the infants, contributed to a developing social bond. The beginnings of parental behavior seem to be present in animals of this age.

Affiliative behavior directed by infants toward an older conspecific (clinging, touching, etc.) are adaptive, since—regardless of whether the stimulus animal is its natural parent—a close relationship with an older animal is necessary for an infant's survival and well-being.

Infant Isolate Compared with Infant Controls

Our infant animals with normal rearing experience behaved quite differently than did those reared in social isolation. At separation the normal infants vocalized more frequently than did isolate infants. At separation normally-reared infants emitted vocalizations which, under natural conditions, would have been likely to reinstate social contact.

Isolate animals, however, exhibited "pathological" responses to separation. Instead of calling frequently as did the control subjects the isolate vocalized comparatively little. Separation, for isolates, increased behaviors directed toward the self. Rocking, floor-clinging, and masturbation increased in these animals when separated from preadolescent cage-mates. Similarly, stereotyped movements and other bizarre behaviors rose during separation. These kinds of responses are of no use to a normal animal, and under natural conditions they would be unlikely to succeed in reinstating social contact.

A high level of self-orientation may indicate a kind of *attachment to self*. An isolate animal is reared in an environment in which the only social stimuli available to it are the moving parts of its own body. By many of the behavioral criteria which we use to measure attachment, isolates *do* show self-attachment (self-touching, self-oriented sexual behavior, digit sucking, etc.). Consider, for example, the behavior of a normal infant when it is frightened. If the infant is with its mother, it will cling to her. When separated from its mother (or a relevant mother substitute) it will cry out, move about and search for its former cage-mate. The isolate has no mother to seek out when frightened, so it increases self-directed behaviors at such times. These increases in self-directed behaviors may, in some fashion, serve the same function as clinging to mother. The primary source of comfort for an isolate, therefore, seems to be the self.

These behaviors may be *internally* adaptive, somehow providing a behavioral buffer from stress-inducing stimuli.

While the greater attachment, as far as the isolate is concerned, seems to be for the self, the absence of an animal with which it has been living for two months does affect its behavior. Our findings lead us to believe, therefore, that the presence of the pre-adolescent cage-mate came to be accepted by the isolate reared infant and that its removal was at least a mildly aversive event. Our descriptive data also indicate that the isolates attached more readily to female than to male preadolescents. Perhaps this occurred because of the females' propensity for maternal-like behavior, but it may also be due to the rough conduct of the males, a potentially aversive stimulus for the controls as well as for the isolate infants.

HETEROSEXUAL VS HOMOSEXUAL PAIRS

Heterosexual pairs, according to our findings, showed a greater degree of protest at separation (more coos, more stereotypies) and more affiliation when together (more visual orientation toward other) than did homosexual pairs. Our data, therefore, provide indications that the beginnings of adult sexual behavior in the preadolescents may affect attachment in dyads of this composition.

In summary, preadolescent monkeys are quite capable of forming attachments to younger animals, although these attachments do not seem to be as strong as the normal infant attachment to the older member of the pair. Such attachments are particularly strong if the pair is heterosexually constituted. However, an infant which exhibits inadequate social behavior will not be as readily accepted as a viable social partner, and when it is removed it will not be as greatly missed.

REFERENCES

BAYSINGER, C., E. M., BRANDT, & G. MITCHELL, 1972. Development of infant rhesus monkeys (*Macaca mulatta*) in their isolation environments. *Primates*, 13: 257–270.

BRANDT, E. M., C. BAYSINGER, & G. MITCHELL, 1972. Separation from rearing environment in mother-reared and isolation-reared rhesus monkeys (*Macaca mulatta*). *International Journal of Psychobiology*, 2: 193–204.

————, & G. MITCHELL, 1973. Pairing preadolescents with infants (*Macaca mulatta*). *Developmental Psychology*, 8: 14–19.

CHAMOVE, A., H. F. HARLOW, & G. MITCHELL, 1967. Sex differences in the infant-directed behavior of preadolescent rhesus monkeys. *Child Development*, 38: 329–335.

ERWIN, J., E. M. BRANDT, & G. MITCHELL, in press. Attachment formation and separation in heterosexually naive pre-adult rhesus monkeys (*Macaca mulatta*)—*Developmental Psychobiology*, 6: 531–538.

————, J. MOBALDI, & G. MITCHELL, 1971. Separation of juvenile rhesus monkeys of the same sex. *Journal of Abnormal Psychology*, 78 (2): 134–139.

HARLOW, H. F. & M. K. HARLOW, 1962. Social deprivation in monkeys. *Scientific American*, 207: 136–146.

HINDE, R. A. & Y. SPENCER-BOOTH, 1971. Effects of brief separation from mother on rhesus monkeys. *Science*, 173: 111–118.

LANCASTER, J. B., 1971. Play mothering: The relations between juvenile females and young infants among free-ranging vervet monkeys (*Cercopithecus aethiops*). *Folia primat.*, 15: 161–182.

MCKINNEY, W. T., S. J. SUOMI, & H. F. HARLOW, 1972. Repetitive peer separations of juvenile-age rhesus monkeys. *Archives of General Psychiatry*, 27: 200–203.

ODOM, B., G. MITCHELL, & D. LINDBURG, 1970. A device to record primate social behavior. *Primates*, 11: 93–96.

REDICAN, W. K. & G. MITCHELL, 1973. A longitudinal study of paternal behavior in adult male rhesus monkeys, 1. Observations on the first dyad. *Develop. Psychol.*, 8: 135–136.

ROSENBLUM, L. A. & I. C. KAUFMAN, 1968. Variations in infant development and response to maternal loss in monkeys. *American Journal of Orthopsychiatry*, 38: 418–426.

SCHLOTTMANN, R. S. & B. SEAY, 1972. Mother-infant separation in the Java monkey (*Macaca irus*). *Jour. Comp. Physiol. Psychol.*, 79: 334–340.

SCOTT, J. P. & F. H. BRONSON, 1964. Experimental exploration of the etepimeletic or care-soliciting behavioral system. In: *Psychobiological Approaches to Social Behavior*, P. H. LEIDERMAN & D. SHAPIRO (eds.), Stanford University Press, Stanford, Calif.

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